

Malek Alhazmi

AI Systems Architect & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine learning engineer and AI systems architect who builds production neural systems end to end: designing transformer architectures, training them from scratch or fine-tuning pretrained backbones, verifying correctness formally, and deploying into regulated and air-gapped environments. As founder of CytherAI, I've shipped systems into Saudi enterprise and government settings that require full data sovereignty, including a ZATCA-certified tax compliance engine in live production and a constrained natural-language CAD system whose outputs are geometrically valid by construction. I also hold a pending US patent for a theorem-proving transformer trained from random initialization on a single consumer laptop, with no GPU cluster and no pretrained weights. Comfortable with mainstream ML tooling (Python, PyTorch, Hugging Face, Linux) as well as resource-constrained optimization on Apple Silicon.

CORE SKILLS

- **AI Systems & Deployment:** Regulated, on-prem, private-cloud, and air-gapped deployment; offline-first architectures with no external dependencies; evaluation-first design for production correctness
- **Verification & Correctness:** Kernel-backed geometric and formal verification (OpenCASCADE B-rep, STEP export); constrained / grammar-based decoding (FSM + degrees-of-freedom masking, constrained beam search)
- **Machine Learning & Deep Learning:** Transformer architecture design and from-scratch implementation (encoder-decoder, multi-head attention, custom output heads); fine-tuning and transfer learning (ModernBERT, LoRA, KV-prefix injection); neural-symbolic reasoning
- **Model Training & Optimization:** Training from random initialization and on pretrained backbones; resource-efficient training on consumer hardware; custom tokenizers; synthetic data generation, curriculum design, class balancing; reproducible fixed-seed pipelines
- **Software Engineering:** Python, Rust, Swift; FastAPI / REST backends and Three.js front-ends; test-driven development (pytest, Vitest, Playwright); MLX, PyTorch, Hugging Face; Linux, Apple Silicon

PROFESSIONAL EXPERIENCE

CytherAI — Founder & AI Systems Architect

Riyadh, Saudi Arabia | 2024 – Present

- Lead engineering, research, and commercial direction for all client work, run through gated Discovery, Architecture, Build, and Deploy phases with explicit acceptance criteria at each stage.
- Build production systems with no cloud APIs and no external runtime calls, fully on-device: core models are built from scratch, with pretrained backbones used only where they add clear value. Filed a US Provisional Patent (2025) on the underlying from-scratch training method.
- Shipped CytherCAD, a constrained generative-CAD system where invalid states are excluded by decoder constraints and kernel-backed verification, with output valid by construction and exportable as STEP.
- Shipped SijilOS, an event-sourced ledger engine for Saudi enterprise tax compliance, to ZATCA Phase 1 certification (April 2025). Live in beta across two production environments, offline-first with no core cloud dependency.
- Based in Riyadh, working with regional and international clients who need sovereign, regulated, or constrained-environment deployments.

SELECTED PROJECTS

CytherCAD — Constrained Natural-Language CAD

Generates parametric CAD programs from natural language that are geometrically valid by construction. An FSM- and degrees-of-freedom-constrained decoder sits over an OpenCASCADE B-rep kernel, so invalid output isn't just filtered, it's structurally impossible. Custom encoder-decoder transformer with a ModernBERT encoder reimplemented in MLX. The full stack, from language understanding to constrained decoding, kernel verification, and STEP export, runs locally on a laptop with no cloud or GPU.

Neural Theorem Prover — Zero-Dependency Research System (Patent Pending)

Custom transformer for automated mathematical proof generation, trained from random initialization with no pretrained weights and no external runtime dependencies. Training data is algorithmically generated and formally verified, and the whole system is reproducible from source. Subject of a pending US patent.

EDUCATION & CERTIFICATIONS

- B.Sc. Computer Science — Jazan University, 2024
- CompTIA Security+ — issued April 2024, valid through April 2027

LANGUAGES

Arabic — Native · English — Full Professional Proficiency